

JAIRAM ALLURI

Dallas, TX · (469) 237-1808 · allurijairam@gmail.com · [LinkedIn](#) · github.com/allurijairam · [Portfolio](#)

SUMMARY

Machine Learning Engineer with 4+ years of production experience across computer vision, Natural Language Processing (NLP), and Large Language Model (LLM) systems. Built and deployed end-to-end ML pipelines on AWS and Azure — from custom Vision Transformer (ViT) architectures and edge ML on embedded hardware to Retrieval-Augmented Generation (RAG) systems with hybrid retrieval. Experienced working at scale on systems handling 600+ TB of data and hundreds of thousands of users. Reduced Azure infrastructure costs by 80% and preprocessing latency by 10× at Honeywell; holds a pending USPTO patent for a novel 3D vision preprocessing technique.

TECHNICAL SKILLS

Languages: Python, SQL

LLM & GenAI: LangChain, Retrieval-Augmented Generation (RAG), RAG Fusion, Multi-Query Retrieval, Prompt Engineering, LLM-as-judge, DeepEval, BM25, Embeddings, Semantic Search, Vector Databases (ChromaDB, FAISS), HuggingFace Transformers

ML Frameworks & Models: PyTorch, TensorFlow, Scikit-learn, BERT, Vision Transformer (ViT), LLaMA, GPT-2, YOLO, OpenCV

MLOps & Cloud: AWS (EC2, Bedrock, S3), Azure (Auto-scaling, Spot Instances), Docker, Flask, REST API, MLflow, CI/CD, GitHub Actions

Techniques: Feature Engineering, Model Quantization, Seq2Seq, Attention Mechanisms, Positional Encodings, Custom Tokenization, Edge ML, Model Monitoring, Experiment Tracking

WORK EXPERIENCE

Machine Learning Engineer | **Honeywell** | Richardson, TX | May 2023 – Present

- Contributed to a large-scale building automation platform processing 600+ TB of sensor and telemetry data across hundreds of thousands of end users — built and maintained ML pipelines for real-time inference at production scale.
- Developed a 3D computer vision system using PyTorch and OpenCV with a custom Vision Transformer (ViT) and specialized positional encodings for structural defect classification in industrial components; the preprocessing technique is pending USPTO patent.
- Reduced preprocessing latency by 10× by re-architecting the Azure pipeline for parallel execution, enabling real-time inference on production workloads.
- Cut Azure cloud spend by 80% through workload-specific instance selection, database configuration tuning, and migrating batch jobs to spot instances.
- Developed and deployed TensorFlow edge ML models for real-time gas concentration prediction on resource-constrained embedded hardware using fixed-point arithmetic with minimal CPU overhead.
- Collaborated with firmware and systems engineering teams to integrate ML inference into embedded deployments, reducing integration cycles from weeks to days.

Machine Learning Engineer | **Straive** | Chennai, India | June 2021 – August 2022

Scientific and academic publishing services firm

- Built a PyTorch deep learning Optical Character Recognition (OCR) pipeline for automated PDF text extraction, reducing per-document processing cost by 90% and eliminating the manual extraction workflow entirely.
- Deployed a YOLO-based object detection system to automate data labeling, cutting manual annotation time by 70% across the team.

- Built a Transformer-based NLP triage pipeline to classify and prioritize customer service queries by urgency, improving resolution speed by 70%.
- Reduced monthly Azure cloud spend by 80% by migrating infrastructure to auto-scaling groups and spot instances, working directly with the DevOps team to validate production rollout.

PROJECTS

Academic Knowledge Assistant — Hybrid RAG Pipeline

LangChain, ChromaDB, BM25, AWS Bedrock, Flask, Docker, Python

- Built a domain-specific Q&A system where students upload course notes and PDFs and receive grounded answers with source citations rather than generic LLM responses.
- Implemented hybrid retrieval combining ChromaDB dense vector search with BM25 sparse keyword search, fused via Reciprocal Rank Fusion (RRF) — improved recall ~30% on technical terminology vs. pure semantic search.
- Added Multi-Query Retrieval to decompose complex questions into sub-queries, improving answer completeness on topics spanning multiple document sections.
- Designed TTL-based session memory (10-min expiry) to maintain conversation context across follow-up questions while automatically purging user data on session end.

→ github.com/allurijairam/rag_lambda_app · [Live Demo](#)

GPT-2 Language Model Built from Scratch

PyTorch, Custom Tokenizer, Transformer Architecture

- Implemented the full GPT-2 decoder-only architecture from scratch in PyTorch: Masked Multi-Head Self-Attention, Feed-Forward Networks, Sinusoidal Positional Encodings, Layer Normalization, and Residual Connections — no HuggingFace or external transformer libraries.
- Built a character-level tokenizer from scratch; trained on a 1M+ token corpus with gradient clipping and residual scaling for stable training across deep transformer blocks.

→ [Technical write-up on Medium](#)

Deep Learning Autocorrect System

PyTorch, Seq2Seq, FastText, 1D-CNN, Flask

- Built a Seq2Seq Encoder-Decoder model with Attention and 1D-CNN layers for real-time sentence-level text correction, handling context-dependent errors that traditional spell-checkers miss.
- Used FastText embeddings for better generalization on rare and misspelled inputs; deployed as a Flask app with sub-100ms end-to-end inference latency.

→ [Technical write-up on Medium](#)

EDUCATION

M.S. in Business Analytics — Data Science Cohort | University of Texas at Dallas | Richardson, TX Aug 2022 – May 2024

GPA: 3.91/4.0 | Dean's Excellence New Student Cohort Scholarship | Ranked #1 in AI/ML and Statistics courses

Relevant coursework: Machine Learning, Deep Learning, Statistical Modeling, Natural Language Processing

PATENTS & RECOGNITION

- **Patent Pending (USPTO)** — Novel data-preprocessing technique for 3D structural pattern classification in industrial computer vision systems, Honeywell International, 2024.
- **Dean's Excellence Scholarship**, University of Texas at Dallas, 2022 — awarded for exceptional academic performance at entry.